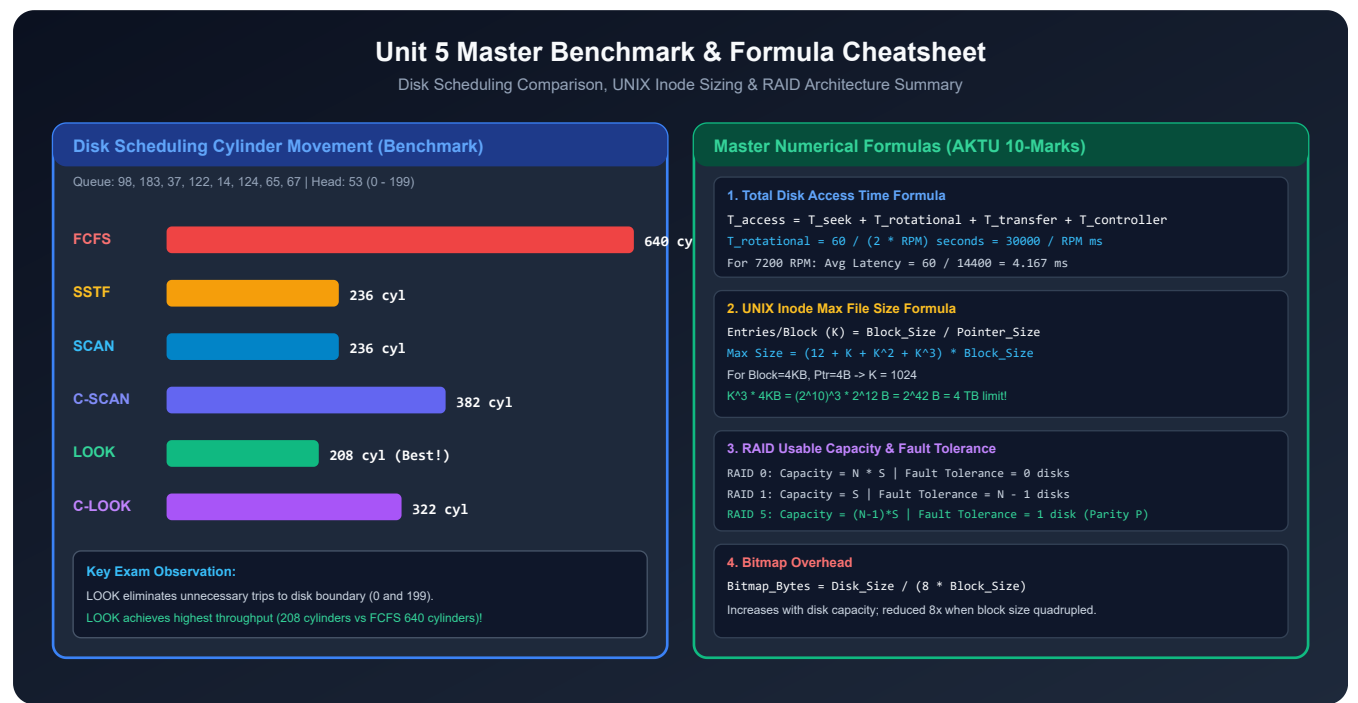


Module 14: Unit 5 AKTU PYQs & Interview Cheatsheet

1. Disk Scheduling Benchmark & Formulas



2. Solved 10-Mark Numerical: Disk Scheduling

Queue: 98, 183, 37, 122, 14, 124, 65, 67 | Initial Head: 53 | Range: 0 to 199

Algorithm	Head Movement Trace Path	Total Seek Count
FCFS	53 → 98 → 183 → 37 → 122 → 14 → 124 → 65 → 67	640 cylinders
SSTF	53 → 65 → 67 → 37 → 14 → 98 → 122 → 124 → 183	236 cylinders
SCAN	53 → 65 → 67 → 98 → 122 → 124 → 183 → 199 → 37 → 14	236 cylinders
C-SCAN	53 → 199 → 0 → 14 → 37	382 cylinders
LOOK	53 → 183 (reverses at max request) → 14	208 cylinders (Best!)
C-LOOK	53 → 183 → 14 → 37	322 cylinders

3. Solved Numerical: UNIX Inode Capacity Derivation

Problem: Block Size = 4 KB, Pointer = 4 B. Inode: 12 Direct, 1 Single, 1 Double, 1 Triple Indirect.

- Pointers per block $K = 4096 / 4 = \mathbf{1024 \text{ pointers } (2^{10})}$.
- Direct = $12 \times 4 \text{ KB} = \mathbf{48 \text{ KB}}$.
- Single Indirect = $1024 \times 4 \text{ KB} = \mathbf{4 \text{ MB}}$.
- Double Indirect = $(1024)^2 \times 4 \text{ KB} = \mathbf{4 \text{ GB}}$.

5. Triple Indirect = $(1024)^3 \times 4 \text{ KB} = 4 \text{ TB}$.

Total Max File Size $\approx 4 \text{ TB}$.

4. Top Core Systems Interview Q&A

Q: Why do SSDs not use SCAN or Elevator scheduling?

A: SSDs use solid-state flash memory without mechanical moving heads or rotating platters. Rotational latency and seek time are 0. Every block has identical $O(1)$ access latency. Hence, SCAN provides zero benefit and adds unnecessary scheduling overhead. Modern SSDs use NVMe FIFO / multi-queue schedulers.